EMMANUEL ADELOJU

PhD Candidate, Learning Literacies and Technology
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Vision for my future lab

My future lab will study sensemaking, that is, how people and AI systems learn by reconciling new data with what they already know, across multiple data modalities (perceptual and informational) and across the boundary between human and machine cognition. My lab will work at the intersection of the learning sciences and AI evaluation, with a focus on K–12 science students and large language and multimodal models. Our guiding question will be: *how do people and AI systems make sense of the world through data*?

Education

2023 – ABD **PhD, Learning Literacies and Technology**
Advisor: Prof. Michelle Jordan | Prof. Punya Mishra
Arizona State University
Dissertation Research: Developing Teachers' Knowledge to Support Data Sensemaking Instruction with GenAI
Research Areas: Data Science Education, Teacher Knowledge

2021 – 2022 **MSc, Cell Biology and Genetics (Cell & Molecular Biology)**
University of Lagos

2018 – 2019 **Postgraduate Diploma in Education**
National Teachers Institute Kaduna, Nigeria

2012 – 2016 **BSc, Biochemistry**
Osun State University

Research Statement

I am a fourth-year PhD candidate (defense scheduled for April 2027) in Learning Literacies and Technology at Arizona State University, co-advised by Dr. Michelle Jordan and Dr. Punya Mishra. My research focuses on *Human-AI Interaction* and *Data Science Education*, specifically, how students and teachers collaborate with Large Language Models (LLM)-based tools during data sensemaking. My dissertation explores how teachers develop knowledge and make sense of scientific data in collaboration with LLM-based tools. Because I'm particularly interested in the intersection of AI capabilities and human learning, I'm also conducting research on what responsible AI looks like in practice, as it relates to AI safety/risks for students.

Dissertation Committee

Michelle Jordan, PhD (Chair),	Arizona State University
Punya Mishra, PhD (Member),	Arizona State University
Ying-Chih Chen, PhD (Member),	Arizona State University

Research & Tools

Skills: Mixed methods, Machine learning, Statistical analysis, Data science
Tools: Figma, IBM SPSS, R, Python (& associated libraries)

Teaching Experience

Semester	Course	Role
Spring 2025	Nature and Society: An Introduction to Environmental Studies	Instructor
Spring 2025	Introduction to Data Analysis	Teaching Asst.
2009 – 2023	A-Level Biology, Chemistry, Mathematics and Physics	Instructor

Awards

Apr 2026 CVPR Conference 2026 Broadening Participation Scholarship Award
Jan 2026 Mary Lou Fulton College Travel Grant
Jan 2026 College Dissertation Research Grant
Apr 2025 Graduate Student Government Graduate Support Award
Feb 2025 Mary Lou Fulton Endowment Scholarship

Media Coverage

June 2026 Designing AI responsibly for youth starts with systems design. https://education.asu.edu/about/news/essay-designing-ai-responsibly-youth-starts-systems-design

Professional & Academic Services

2026 Reviewer, *CHI 2027*
2026 Associate Journal Editor, *Current Issues in Education*
2026 Reviewer, *Communications of the Association for Information Systems*
2026 Reviewer, *Doctoral Council of the Teachers College Conference*
2026 Reviewer, Graduate Student Research Symposium – *99th NARST Annual International Conference*
2025 Reviewer, *Current Issues in Education*

Volunteering & Mentorship

Fall 2025 – 2026 Learning Literacies and Technology (PhD) Program Officer,
Fall 2024 – 2025 Co-founder of GradMeetup,
Fall 2023 - Present Science mentor, Alhambra Elementary School District Middle Sonoran Photovoltaics Lab students

PUBLICATIONS

Peer Reviewed Articles

Suloff, S., Jordan, M. E., Weinberg, A. E., Coleman, J., **Adeloju, E.**, & Oster, N. (2026). From Knowledge to Informed Action: Action-Oriented Pedagogies in Climate and Safety Science Education. Transforming Schools and Communities to Address the Climate Crisis, 249.

Adeloju, E., Mishra, P., McCaleb, L., & Oster, N. (2026). Designing in uncertainty: GenAI and creative agency in a graduate seminar. *Journal of Instructional Design and Technology.*

Oladele, J., Oyewole, O., Oyeleke, M., Adewale, O., & **Adeloju, E.** (2019). Annona muricata Attenuates Cadmium-Induced Oxidative Stress and Renal Toxicity in Wistar Rats. *Journal of Bioscience and Applied Research*, 5(4), 543-550.

Refereed Conference Proceedings

Adeloju, E., McCaleb, L., Jongewaard, R, Oster, N., & Mishra, P. (2026, Feb 3-4). Socio-Emotional Learning in AI K-12 Guidance and Policy Documents: A Gap Analysis. In Craig, S. D., Arner, T., & McNamara, D. S. (Eds.), *Proceedings of the Learning Engineering Research Network Convening (LERN 2026): From Insights to Implementation, Learning Engineering in Action*. LERN Convention Proceedings. https://doi.org/10.59668/2551.25384

Adeloju, E., McCaleb, L., Oster, N., Jongewaard, R., & Mishra, P. (2026). Education by Design: Navigating Wicked Problems with Generative AI and Collaborative Learning. Proceedings of Society for Information Technology & Teacher Education International Conference. (p. 2050-2055)

Jongewaard, R., McCaleb, L., Oster, N., **Adeloju, E.**, & Mishra, P. (2026). Social and Emotional Dimensions of Generative AI Use. Proceedings of the Learning Engineering Research Network Convening (LERN 2026). (p. 305-306) https://doi.org/10.59668/2551.25165

Oster, N., Jongewaard, R., McCaleb, L., **Adeloju, E.**, & Mishra, P. (2026). The Psychological Other: Student Relationships with GenAI, an Exploratory Study. Proceedings of Society for Information Technology & Teacher Education International Conference. (p. 873-878)

Meza-Torres, C., Jordan, M., Zuiker, S., Jongewaard, R., **Adeloju, E.,** & Spreitzer, K. (2024). Examining teacher supports for visibility, believability, and meaningfulness in place-based citizen science. In Lindgren, R., Asino, T. I., Kyza, E. A., Looi, C. K., Keifert, D. T., & Suárez, E. (Eds.), *Proceedings of the 18th International Conference of the Learning Sciences - ICLS 2024* (pp. 2409-2410). International Society of the Learning Sciences. https://doi.org/10.22318/icls2024.920608

Book Chapters

Suloff, S., Jordan, M., Weinberg, A., Coleman, J., **Adeloju, E.,** & Oster, N. (In press). From knowledge to informed action: Action-Oriented Pedagogies in climate and safety science education. In F. Canady, A. Panos, S. Evans, R. Beach, & J. Castek (Eds). *Addressing the climate crisis for transforming schools and communities* (p. 249). Bloomsbury Academic, Inc.

Ajayi, E. I., Molehin, O. R., Ajayi, O. O., **Adeloju, E.,** & Oladele, J. O. (2023). Application of chitosan-coated foods, fruits and vegetables on inflammation in metabesity. In *Next Generation Nanochitosan* (pp. 431-446). Academic Press.

Under Review & In Preparation

Adeloju, E., & Jordan, M. E. (2026). *Who Makes Sense of the Data? Roles and Moves in Teacher-LLM Collaboration* [Manuscript submitted for publication]. Mary Lou Fulton College for Teaching and Learning Innovation, Arizona State University.

Adeloju, E. (2026). *Data Sensemaking: A Theoretical Framework for Scientific Explanation and Knowledge Generation* [Manuscript submitted for publication]. Mary Lou Fulton College for Teaching and Learning Innovation, Arizona State University.

Adeloju, E., McCaleb, L., Jongewaard, R., Oster, N., & Mishra, P. (2026). *From Policy to Design: Uncovering Socioemotional Risks in Generative AI Systems for Education* [Manuscript in preparation]. Mary Lou Fulton College for Teaching and Learning Innovation, Arizona State University.

Refereed Presentations, Posters and Workshops

Adeloju, E., McCaleb, L., Jongewaard, R., Oster, N., & Mishra, P. (2026, June 4). *From policy to design: Uncovering socioemotional risks in generative AI systems for education* [Lightning talk]. Humans of Generative AI (HuG) Workshop, CVPR 2026, Denver, CO, United States. https://humansofgenerativeai.github.io/

Jordan, M., **Adeloju, E.**, Weinberg, A. Oster, N., & Anayatova., D. (2026, April 19-22). *Opportunities for Student Agentic Action in K-12 Sustainability Education: A Systematic Review* [Conference presentation]. Ninety-ninth Annual NARST International Conference, Seattle, Washington, United States.

Adeloju, E., Jordan, M., Suloff, S., Oster, N., & Weinberg, A. (2026, April 19-22). *In-Service Teachers Cultivating Student Agency through Safety Science: An Evaluation Report* [Conference presentation]. Ninety-ninth Annual NARST International Conference, Seattle, Washington, United States.

Adeloju, E., McCaleb, L., Jongewaard, R, Oster, N., & Mishra, P. (2026, Feb 20). Socio-Emotional Learning in AI K-12 Guidance and Policy Documents: A Gap Analysis [Conference presentation]. *The Doctoral Council at the Teachers College Conference*, Tempe, AZ, United States.

Adeloju, E., McCaleb, L., Jongewaard, R, Oster, N., & Mishra, P. (2026, Feb 3-4). *Socio-Emotional Learning in AI K-12 Guidance and Policy Documents: A Gap Analysis* [Conference presentation]. Learning Engineering Research Network Convening (LERN 2026): From Insights to Implementation, Learning Engineering in Action, Tempe, Arizona, United States

Adeloju, E. (2026, Feb 3-4). *Integrating AI into Datasensemaking: Teachers' Learning and Pedagogical Reflections* [Poster presentation]. Learning Engineering Research Network Convening (LERN 2026): From Insights to Implementation, Learning Engineering in Action, Tempe, Arizona, United States

Adeloju, E., Jordan, M. E., Weinberg, A. E., Oster, N., & Anayatova, D. (2025, October 30). *Consequential action in K-12 sustainability education: A systematic review* [Conference Presentation]. North American Association for Environmental Education (NAAEE) Research Symposium, online.

Oster, N., Nichols, J., Jordan, M. E., Suloff, S., Weinberg, A. E., **Adeloju, E.,** & Anayatova, D. (2025, October 30). *Interdisciplinary practices for imagining climate futures in "Future City" projects* [Conference Presentation]. North American Association for Environmental Education (NAAEE) Conference online.

Jordan, M., Meza-Torres, C., Jongewaard, R., Zuiker S., & **Adeloju, E**. (2025, March 22-26). *Building a regional network of school-based citizen scientists for sustainable energy futures* [Conference presentation]. In Silova, I. (chair), Education for Planetary Futures (part 2): Disrupting, Reimagining, and Co-Designing Pedagogies on the Edge symposium. Comparative and International Education Society (CIES) Conference Chicago, IL, United States.

Oster, N., Anayatova, D., Weinberg, A., Jordan, M., & **Adeloju, E.** (2025, March 23). *Preservice teachers' visions for Action-Oriented Pedagogies for sustainability education.* In Silova, I. (Chair), Education for Planetary Futures (Part 2): Disrupting, Reimagining, and Co-Designing Pedagogies on the Edge symposium. Presented at the CIES 2025 Conference, March 22-26, 2025, Chicago, IL.

Jordan, M., Meza-Torres, C., Jongewaard, R., Zuiker S., & **Adeloju, E**. (2025, March 26-29). *Building a regional network of school-based citizen scientists for sustainable energy futures* [Conference presentation]. National Science Teachers Association (NSTA) Conference, Philadelphia, PA, United States.

Jordan, M., Jongewaard, R., Zuiker, S., Meza-Torres, C., & **Adeloju, E.** (March 2025). *Building a regional network of school-based citizen scientists for sustainable energy futures.* Comparative and International Education Society (CIES) Annual Conference. Chicago, IL.

Jordan, M., Jongewaard, R., Zuiker, S., Meza-Torres, C., & **Adeloju, E.** (2025, March 23). *Learning and Contributing to Sustainable Futures: A regional network of school-based citizen scientists*. In Silova, I. (Chair), Education for Planetary Futures (Part 2): Disrupting, Reimagining, and Co-Designing Pedagogies on the Edge symposium. Presented at the CIES 2025 Conference, March 22-26, 2025, Chicago, IL.

Suloff, S., Adeloju, E., Jordan, M., Coleman, J., Coates, M., Mendoza, J., Buckner, E., Weinberg, A., & Oster, N. (2025, March 29). *Safety Science and Sustainability: Online resources to inspire student action.* Presented at the National Science Teachers Association (NSTA) Conference, Philadelphia, PA.

Jordan, M., Rapkiewcz, J., Ramos, M., Coates, M., & **Adeloju, E.** (2025, March 29). *Creating a regional network of K-12 community scientists for sustainable Futures.* Presented at the National Science Teachers Association (NSTA) Conference, Philadelphia, PA.

Jordan, M., Weinberg, A., **Adeloju, E**., Oster, N., & Coleman, J. (2025, March 26-29). *Safety Science and Sustainability: Online resources to inspire student action* [[Conference presentation]. National Science Teachers Association (NSTA) Conference, Philadelphia, PA, United States.

Jordan, M., Weinberg, A., **Adeloju, E**., & Oster, N. (2025, March 7). *Formative Assessment Report #3: Xplorlabs Educator Fellowship Community of Practice (CoP)*. Arizona State University, Tempe, AZ.

Meza-Torres, C., & **Adeloju, E**. (2025, April 23-27). *Reimagining School-based Citizen Science for Sustainable Futures: K-12 teachers fostering visibility, believability, and meaningfulness* [Conference presentation]. In Weinberg, A. (chair), Education for Imagining and Advancing Preferred Planetary Futures. AERA Conference Denver, CO, United States.

Adeloju, E., Jordan, M., & Meza-Torres, C. (2025, April 22). *Machine Learning for Validating Citizen Science Data: A Case Study with High School Students in Agri-Photovoltaics Project in the*

Sonoran Desert Region [Poster presentation]. Institute for Social Science Research, Arizona State University, Tempe, AZ, United States.

Adeloju, E., Jordan, M., & Meza-Torres, C. (2025, February 21). *Machine Learning for Validating Citizen Science Data: A Case Study with High School Students in Agri-Photovoltaics Project in the Sonoran Desert Region* [Conference presentation]. *The Doctoral Council at the Teachers College (TCDC) Conference*, Tempe, AZ, United States.

Oster, N., **Adeloju, E.,** Anayatova, D., Jordan, M. E., Weinberg, A. E. (January 2025). *Advancing education for planetary futures through action-oriented pedagogies: Insights from two studies.* Learning Planet Festival. Virtual. https://www.learning-planet.org/event/advancingeducation-for-planetary-futures-through-action-oriented-pedagogies-insights-from-twostudies/

Suloff, S., Jordan, M. E., Weinberg, A. E., Oster, N., **Adeloju, E.** (January 2025). *Co-creating preferred futures: The Xplorlabs Educator Fellowship experience.* Learning Planet Festival. Virtual. https://www.learning-planet.org/event/co-creating-preferred-futures-the-xplorlabseducator-fellowship-experience/

Adeloju, E., Jordan, M., & Weinberg, A. (2024, December 7). *A case study of AOP Safety Science practice of a youth at the Action Oriented Pedagogy Community of Practice Co-design workshop for Arizona teachers* [Workshop presentation]. In, Jordan, M. & Weinberg, A. (Principal Investigators), Xplorlabs Teacher Workshop. Arizona State University, Tempe, AZ.Available at: https://docs.google.com/presentation/d/1QdAoxRLcf4q06N5SFJZu-ugPf-H4pYwt8rDoS2i0Qy8/edit#slide=id.p

Jordan, M., Weinberg, A., **Adeloju, E**., & Oster, N. (2024, November 22). *Formative Assessment Report #2: Xplorlabs Educator Fellowship ULRI Annual Research Symposium* [Conference presentation]. Atlanta, GA. Available at https://ulnfp.sharepoint.com/:p:/r/sites/XplorlabsEducatorFellowship/Shared%20Documents/Research%20Symposium/ULRI%202024%20Annual%20Research%20Symposium%20Survey%20Results.pptx?d=wad4a2006ce1d473587c6fedea5957287&csf=1&web=1&e=jsKsMf

Adeloju, E., Oster, N., Curtiss, B., Hubert, L., Jordan, M., & Weinberg, A. (2024, November 7-8). *Empowering Student Action: Integrating Safety Science and Sustainability with Xplorlabs*. Arizona Science Teachers Association (ASTA) 70th Annual Conference, Mesa, AZ, United State. https://azsta.org/events/2024-conference/

Jordan, M., Weinberg, A., **Adeloju, E**., & Oster, N. (2024, August 30). *Formative Assessment Report #1: Xplorlabs Educator Fellowship Summit* [Conference presentation]. Arizona State University, Tempe, AZ. Available at https://ulnfp.sharepoint.com/:p:/r/sites/XplorlabsEducatorFellowship/Shared%20Documents/July%20Workshops/Formative%20Evaluation%20%231_%202024%20Fellowship%20Summit%20-%20Xplorlabs%2024_25.pptx?d=w37d5846ad90f489da9deb5f4a51c1e6a&csf=1&web=1&e=fTd3is

Adeloju, E., Jordan, M., Meza-Torres, C., & Jongewaard, R. (2024, June 28). *Introduction to PN junction chemistry for Solar panels. Sonoran Photovoltaics Lab Teacher workshop* [Workshop presentation]. In, Jordan, M. (Principal Investigator), SPV Lab summer teacher workshop. Arizona State University, Tempe, AZ. Available at https://docs.google.com/presentation/d/1kcv3Bq4ui86tFR6_iF49g1PPLMYBSUJMaq_Mx7gJGqM/edit#slide=id.p

Adeloju, E., & Jordan, M. (2024, February 9). *Impact of storytelling on the academic grades and interest of 10th grade chemistry students*. *The Doctoral Council at the Teachers College (TCDC) Conference*, Tempe, AZ, United States.

Suloff, S., Oster, N., **Adeloju, E.,** Curtiss, B., Hubert, L., Jordan, M., & Weinberg, A., Colton, J., Mannis, K., McDonald, M., Avila, A., Enriquez, I., & Van Lith, L. (2024, January 24). Co-Creating Preferred Futures: The Xplorlabs Educator Fellowship Experience. Learning Planet Festival. Tempe, AZ, United States. https://www.learning-planet.org/event/co-creating-preferred-futures-the-xplorlabs-educator-fellowship-experience/

Meza-Torres, C., Jordan, M., Zuiker, S., Jongewaard, R., & **Adeloju, E.** (2023). *Fostering meaningfulness, visibility, and believability in youth engagement with place-based citizen science data.* Scholarly Consortium for Innovative Psychology in Education (SCIPIE). Portland, Oregon

Conference Attendance

2024 Annual UL Research Institutes Research Symposium: Protecting Human Health and Safety in the Modern World. Atlanta, GA, United States. (Sept 30th – Oct 2nd, 2024)

2024 NARST Annual International Conference: Science education for the rest of us. Denver, CO, United States. (March 17-20). https://narst.org/conferences/2024-annual-conference

Panel / Invited Talks

Adeloju, E. (2026, March 25). Consideration for AI in Research [Guest Speaker]. In Dr. Jordan, M. Graduate Science Methods, AZ, United States.

Adeloju, E. (2025, November 17). Introduction to LLM for Datasensemaking [Guest Speaker]. ASU Preparatory Academy, Casa Grande, AZ, United States.

Enrique, I., Avila, A., Sosa, T., **Adeloju, E.**, Buckner, E, Mendoza, J., Ankrum, J., & Coates, M., Orr, D. (August 29, 2024). The Power of Youth: "Tell me, what is it you plan to do with your one wild and precious life? [Panel session]. In, Pandya, R. (Chairperson), Education and Democracy in a hotter time: A wake-up call to climate action symposium panel. ASU Memorial Union, Tempe, AZ. Video Available at: https://www.youtube.com/watch?v=q9KlnuCoOaA

Blogposts & Public Scholarship

Adeloju, E. (2026, March 2). *What AI sycophancy means for K–12 students*. Emmanuel Adeloju. https://prodholly.github.io/EmmanuelAdeloju/blog/2026/ai-sycophancy-students/

Adeloju, E. (2026, February 1). *Prompt engineering framework for data sensemaking*. Emmanuel Adeloju. https://prodholly.github.io/EmmanuelAdeloju/blog/2026/prompt-framework/

Adeloju, E. (2026, February 1). *Prompting techniques for science data analysis and sensemaking*. Emmanuel Adeloju. https://prodholly.github.io/EmmanuelAdeloju/blog/2026/prompt-techniques/

Adeloju, E. (2026, February 1). *Intro to large language models*. Emmanuel Adeloju. https://prodholly.github.io/EmmanuelAdeloju/blog/2026/intro-to-llm/

Adeloju, E., Suloff, S., Weinberg, A., Ambrose, G. (2024, November 19). From Safety Science to Sustainability: Insights from Xplorlabs Educator Fellows at the ULRI Annual Research Symposium. *Education for Planetary Futures Substack*. Available at https://edforplanetaryfutures.substack.com/p/from-safety-science-to-sustainability

Culminating Products

Mitchell, S. & **Adeloju, E**. & Henderson, B. (2024). ICAP teacher observational protocol for college science classes [Pilot observational protocol]. Scottsdale Community College, Scottsdale, AZ.